



Semantic Clustering and Classification of Food Preparation Protocols

Harsh Vishwakarma(2022205) | Dhairya(2022157)
BTP Advisor: Dr. Ganesh Bagler



Introduction

Food preparation protocols (cooking action verbs like boil, chop, knead, blend) are the fundamental building blocks of recipes, yet no standardized semantic classification of these actions exists. Traditional food classification systems like NOVA focus on ingredients and formulation but ignore how food is actually prepared. This study investigates whether NLP-based embedding models can automatically group 270 unique culinary action verbs into meaningful semantic categories that reflect their role in food processing. We compare seven embedding models SBERT, CookBERT, RecipeBERT, FoodBERT, BioBERT, PubMedBERT, and Doc2Vec using both internal and external clustering metrics, and evaluate them against a manually constructed ground-truth taxonomy of 25–26 protocol categories.

Problem statement and objective

Existing food classification methods lack a standardized framework for categorizing cooking actions based on their semantic role and processing intensity.

This raises key questions:

- Can cooking actions be grouped into meaningful semantic categories?
- Which embedding model best captures these relationships?
- Can we objectively evaluate clustering quality against human understanding?

Dataset Overview

We use RecipeDB, a large-scale dataset containing over 118,000 recipes from diverse cuisines worldwide.

From the “Processes” field, we extracted ~270 unique cooking protocols.

Preprocessing steps:

- Lowercasing and lemmatization
- Removal of non-action tokens
- Deduplication of protocol terms

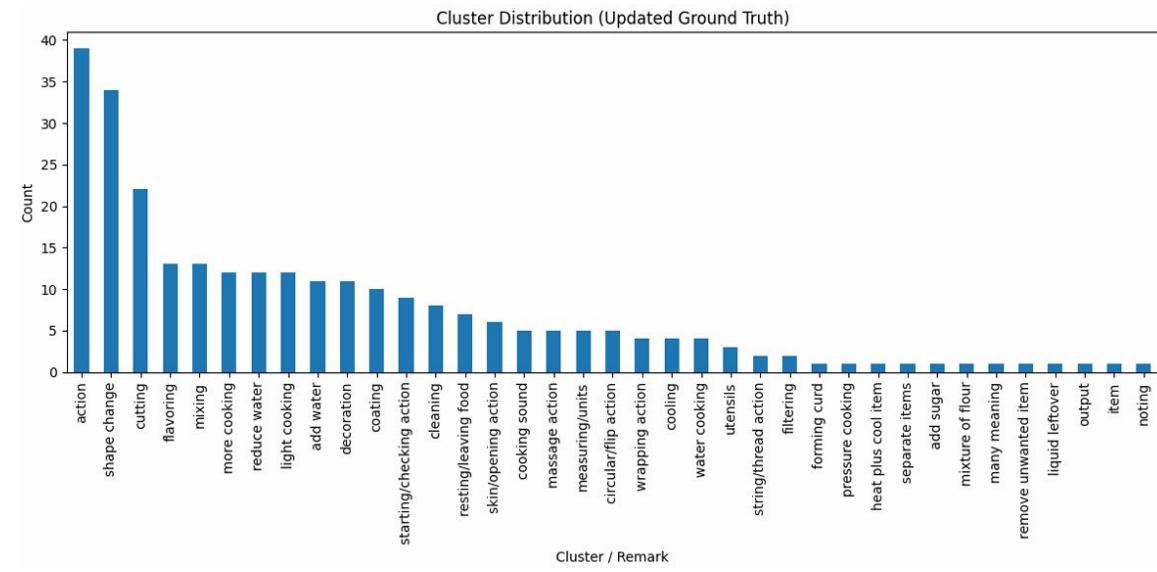
Methodology

A. Protocol Extraction and preprocessing
From the RecipeDB "Processes" field, every cooking-action verb was programmatically extracted and preprocessing steps were applied like Tokenization, Lemmatization, Deduplication and filtering. Result we get is 270 unique culinary protocol terms

B. Ground Truth Construction
To enable external evaluation, each of the 270 protocols was manually labeled using definitions from four authoritative sources: **Oxford English Dictionary**, **Cambridge Dictionary**, **Merriam-Webster**, and the cooking-specific resource **Food.com** (2–3 definitions per protocol). Based on these definitions, protocols were assigned to semantic subcategories:

- *mix, blend, stir, combine, fold* → **Mixing / Combination**
- *boil, steam, fry, bake, roast* → **Thermal Processing**
- *chop, slice, dice, mince* → **Cutting / Separation**
- **+more**

Final taxonomy: **25–26 semantic categories**, each protocol assigned exactly one label. This (protocol, category) labeled dataset served as the ground truth for evaluating clustering performance.



C. Embedding Models

Seven embedding models were used to generate vector representations of the 270 protocol terms. For multi-word protocols (e.g., *stir-fry, pan-sear*), the final embedding was computed as the **mean of individual token vectors**.

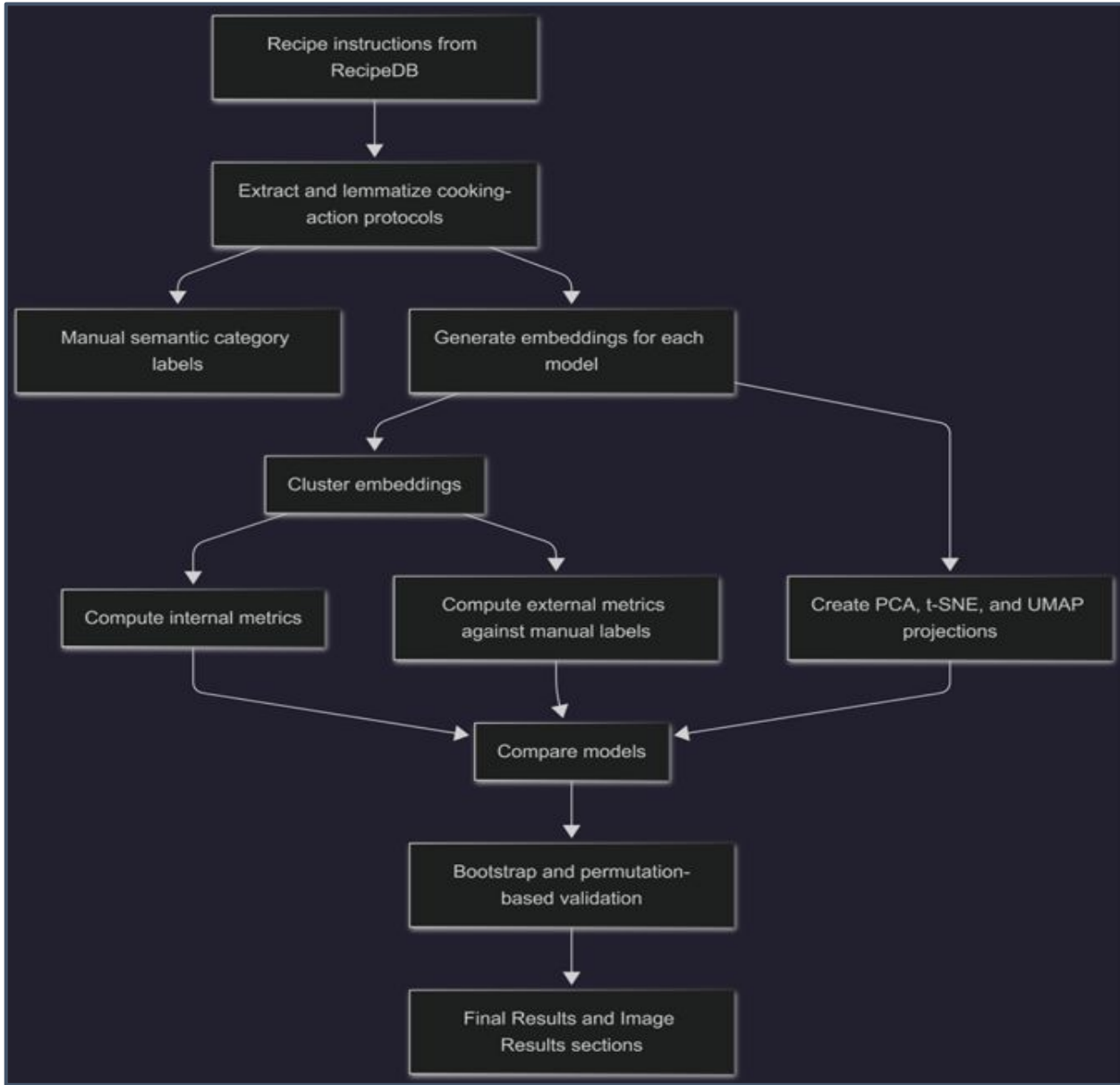
Model	Domain
SBERT	General-purpose sentence transformer
CookBERT	Cooking-domain fine-tuned BERT
RecipeBERT	Recipe-domain fine-tuned BERT
FoodBERT	Food-domain fine-tuned BERT
BioBERT	Biomedical-domain BERT
PubMedBERT	Biomedical literature BERT
Doc2Vec	Paragraph-level embedding (baseline)

To enrich model input, **contextual definitions** from dictionary sources were concatenated with the verb during embedding generation.

D. Clustering & Evaluation

- **Clustering Method: Agglomerative (Hierarchical) Clustering** — builds clusters by repeatedly merging the closest groups
- **Internal Metric: Silhouette Score** — measures how well-separated clusters are geometrically
- **External Metric: Adjusted Rand Index (ARI)** — measures agreement between generated clusters and manual ground-truth labels (primary evaluation metric)
- **Visualizations: PCA** (linear 2D projection), **t-SNE** (nonlinear local structure), **UMAP** (nonlinear global + local manifold structure)

Pipeline Architecture



After constructing a manually labeled ground-truth dataset of cooking protocols, we generated embeddings using multiple models (SBERT, CookBERT, RecipeBERT, BioBERT, PubMedBERT, and Doc2Vec). These embeddings were clustered using agglomerative clustering. The clustering quality was evaluated using Silhouette Score (internal metric) and Adjusted Rand Index (ARI) against the ground truth. Finally, we analyzed the embedding space using PCA, t-SNE, UMAP, and dendrograms to understand both global and local semantic structure.

Clustering and Evaluation

Clustering Method

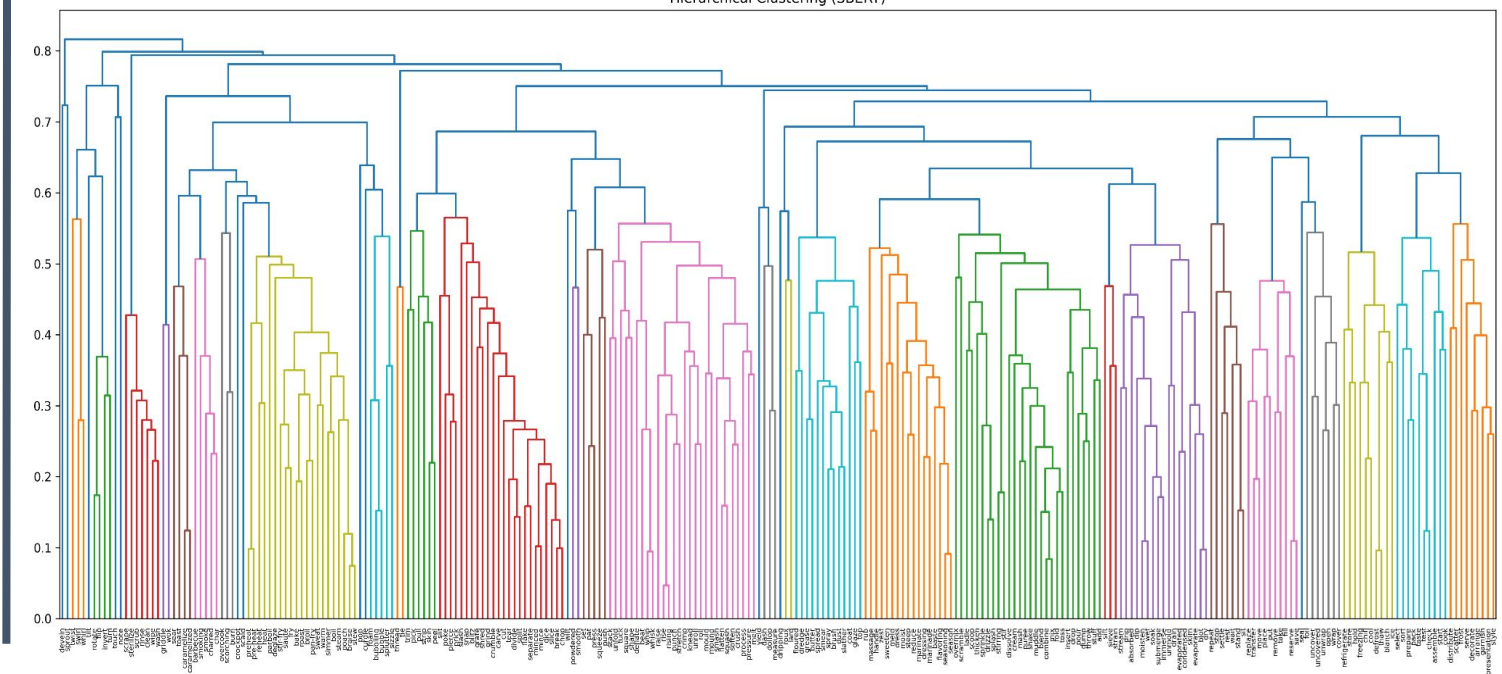
Agglomerative (Hierarchical) Clustering builds clusters by iteratively merging the two closest groups. Applied to embeddings of all 270 protocol terms from each of the 7 models.

Evaluation Metrics

- **Silhouette Score (Internal):** Measures geometric separation how well-separated clusters are in embedding space. Higher = tighter, better-separated clusters. Can be misleading without external validation.
- **Adjusted Rand Index:** Measures agreement between generated clusters and the manually constructed ground-truth semantic categories. Range: -1 to 1. Higher = better alignment with human labels. $ARI \approx 0$ means random alignment.
- **Normalized Mutual Information (NMI)** measures the shared information between predicted clusters and ground-truth labels. Higher NMI indicates better alignment
- **V-measure** evaluates clustering as the balance of homogeneity and completeness. Higher values indicate more consistent and complete clusters.

Visualizations Generated

- PCA — Linear 2D projection; reveals global embedding structure
- t-SNE — Nonlinear projection; emphasizes local neighborhood similarity
- UMAP — Nonlinear; balances local + global manifold structure (clearest results)
- Dendrograms — Hierarchical merge tree; shows semantic grouping quality



RESULTS

Model	Silhouette	ARI	NMI	V-measure
SBERT	0.1655	0.5360	0.7144	0.7140
CookBERT	0.1228	0.2810	0.6220	0.6212
BioBERT	0.0932	0.1678	0.5390	0.5379
RecipeBERT	0.0586	0.1550	0.5491	0.5458
FoodBERT	0.0587	0.1550	0.5491	0.5458
PubMedBERT	0.0212	0.0686	0.4388	0.4242
Doc2Vec	0.6333	-0.0036	0.2286	0.1777

Finding 1 - SBERT is best: Achieved the highest ARI of 0.5360, indicating the strongest agreement with human-labeled taxonomy. SBERT embeddings preserved semantic similarity consistent with expert-defined protocol classes, outperforming all 6 other models.

Finding 2 -Domain-specific models underperform: CookBERT ranked 2nd with ARI 0.2810 — significantly below SBERT. FoodBERT and RecipeBERT scored 0.1550, and PubMedBERT scored worst among transformers (0.0686), confirming that biomedical knowledge does not transfer to culinary tasks.

Finding 3 - Visual separation ≠ Semantic correctness: Doc2Vec had the highest Silhouette Score (0.6333) yet ARI of -0.0036, essentially random. Its UMAP showed dramatic two-cluster separation that was completely misleading. This is the key methodological lesson: internal metrics alone cannot validate semantic quality.

References

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Complete work details

Project GitHub Repositories:

- [Home-Cooked Food Protocol Classification](#)
- [Numerical Model: Nutrient & NOVA Classification](#)
- [Food Pairing Analysis](#)
- <https://github.com/Harshvishwa123/BTP-8TH-SEM>